

TASK 2 – DATABASE INTEGRATION USING SPRING BOOT

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Project Title

Employee Management System – Database Integration using Spring Boot with Flyway

Objective

To integrate a Spring Boot application with a relational database (H2) and manage database schema using Flyway migrations. The project demonstrates automatic table creation and data insertion using version-controlled SQL scripts.

Tech Stack Used

- Java 17
- Spring Boot 3.x
- Spring Data JPA
- H2 Database (In-memory)
- Flyway Migration Tool
- Maven

Project Description

This project extends a CRUD-based Employee Management System by integrating a database and applying Flyway migrations. Instead of manually creating tables, SQL scripts are executed automatically when the application starts.

Database Configuration

```
spring.datasource.url=jdbc:h2:mem:testdb
```

```
spring.datasource.driverClassName=org.h2.Driver
```

```
spring.datasource.username=sa
```

```
spring.datasource.password=
```

```
spring.jpa.hibernate.ddl-auto=update
```

spring.h2.console.enabled=true

spring.flyway.enabled=true

Project Structure

src/main/resources/db/migration/

Migration File 1 – V1__create_employee_table.sql

```
CREATE TABLE employee (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(255),  
    email VARCHAR(255),  
    department VARCHAR(255)  
);
```

Migration File 2 – V2__insert_sample_data.sql

```
INSERT INTO employee (name, email, department)  
VALUES ('Keerthi', 'keerthi@gmail.com', 'CSE');
```

```
INSERT INTO employee (name, email, department)  
VALUES ('Rahul', 'rahul@gmail.com', 'IT');
```

Console Output (Proof of Execution)

Successfully applied 2 migrations to schema "PUBLIC"

Migrating schema "PUBLIC" to version 1

Migrating schema "PUBLIC" to version 2

API Endpoint Tested

GET http://localhost:8080/api/employees

Sample Response

[

```
{
  "id": 1,
  "name": "Keerthi",
  "email": "keerthi@gmail.com",
  "department": "CSE"
}
```

Testing Tool Used

- Thunder Client / Postman

Screenshots to Add

```
PS C:\Users\Keerthi\Downloads\demo\demo> .\demo.cmd spring-boot:run
2026-06-01T22:08:43.156+05:30 INFO 21460 --- [main] com.example.demo.DemoApplication : No active profile set, falling back to 1 default profile: "default"
2026-06-01T22:08:43.548+05:30 INFO 21460 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-06-01T22:08:43.594+05:30 INFO 21460 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 35 ms. Found 1 JPA repository interface.
2026-06-01T22:08:43.883+05:30 INFO 21460 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-06-01T22:08:43.891+05:30 INFO 21460 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-06-01T22:08:43.892+05:30 INFO 21460 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.54]
2026-06-01T22:08:43.928+05:30 INFO 21460 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-06-01T22:08:43.929+05:30 INFO 21460 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 739 ms
2026-06-01T22:08:44.074+05:30 INFO 21460 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-06-01T22:08:44.181+05:30 INFO 21460 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:testdb user=SA
2026-06-01T22:08:44.183+05:30 INFO 21460 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
```

GitHub Repository

<https://github.com/keerthibandari167/employee-crud-api>

Key Learnings

- Spring Boot database integration
- Flyway migration system
- Version-controlled database schema
- REST API with persistent storage

Conclusion

The project successfully demonstrates database integration using Spring Boot and Flyway migrations. It ensures structured schema management and automatic data insertion during application startup.

